

DEXTER FERNANDES

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SUMMARY

Software engineer with 3+ years designing and maintaining **backend software** in **real-time, performance-critical C++**, close to hardware and sensors. Built asynchronous C++17 perception pipelines on Intel edge hardware, raising throughput from 5 to 15 FPS, and NVIDIA DeepStream image and video processing pipelines fusing **multi-sensor** camera and channel data for 24/7 industrial operation. MSc dissertation combined and processed data from **LiDAR**, depth-camera and radiological sensors in a real-time C++17 and ROS2 fusion system on a mobile robot. MSc Robotics, University of Bristol; comfortable owning software end to end -- design, testing, debugging and performance tuning -- as part of a distributed engineering team.

RELEVANT EXPERIENCE

GrowthStage — Technical Screening Engineer (Contract) 05/2026 – PRESENT · REMOTE

- Assess candidate technical depth across perception, SLAM and deployment through hands-on robotics and **computer vision** screens.
- Write and calibrate evaluation rubrics for robotics engineering candidates.

Independent Engineering — Self-Directed Projects (Career Break) 09/2024 – 04/2026 · REMOTE

- Self-directed engineering programme in C++ and real-time perception, including a DeepStream 9.0 multi-stream RTSP pipeline with a custom C++ and CUDA TensorRT decode plugin, detailed under Projects.
- Rebuilt and benchmarked current detector families against production criteria including accuracy, latency, model size and export path, to keep pace with the state of the art in edge perception.

Esbaar — AI Engineer 01/2023 – 08/2023 · MUSCAT, OMAN

- Developed and maintained NVIDIA DeepStream **real-time** video inference pipelines for 24/7 industrial perception, covering camera ingestion, model inference and field diagnostics in harsh desert environments, with a repeatable FP16 optimisation workflow validated against latency, memory and stability thresholds across hardware targets.
- Built an activity classification workstream over 91 **sensor channels** using gradient boosting, random forests and MultiRocket for multivariate time-series classification.
- Improved service **reliability** by hardening deployed services with Dockerised microservices, Kafka messaging, health checks and safe restarts, cutting live video pipeline crashes by 67% and enabling recovery without site visits, using timeouts, retries and message validation with structured logs for root cause analysis under load.
- Worked test-first on the Python inference services, writing failing pytest cases against detection and health-check contracts before implementing them, keeping interfaces stable as the pipeline changed underneath.
- Built a Django and Bootstrap internal dashboard surfacing inference outputs, system health and performance metrics, designing the **user interface**, views, templates and forms that let operators inspect live camera feeds and inference results without engineer involvement.
- Led field validation on BP oil rig deployments, diagnosing model failures under lighting, motion and environmental stressors, and **documented** fixes and recovery procedures for the engineers supporting fielded systems.

LivNSense Technologies — Lead Software Engineer, AI 05/2020 – 12/2022 · BENGALURU, INDIA

- Built asynchronous C++17 perception pipelines on OpenVINO and Intel Movidius VPUs, raising throughput from 5 to 15 FPS while holding deterministic real-time inference on **embedded** Intel edge hardware.
- Overlapped pre-processing, inference and post-processing stages in the C++ pipeline to reduce end-to-end latency and improve utilisation on Intel GPU targets.
- Profiled C++ inference and pre/post-processing hotspots with Intel VTune Profiler, driving targeted performance fixes in the CNN inference path, and integrated Boost C++ libraries into production pipelines, resolving dependency and build issues to keep deployments reproducible.
- Drove the C++ inference path test-first with gtest, specifying pre-processing, inference and post-processing behaviour before implementation, which made the async refactor from 5 to 15 FPS safe to attempt.
- Evaluated edge versus cloud inference trade-offs across **complex systems integrating software, hardware and data**, minimising bandwidth, latency and operational cost while holding **safety-critical** performance targets, and designed the PostgreSQL schema routing real-time incidents into downstream analytics.
- Led four engineers building a two-stage detector feeding helmet-colour classification for construction safety in Texas, cutting false positives by 33% within latency and power budgets, and worked with hardware, software and QA engineers across the **full development lifecycle** to integrate perception modules into deployed systems.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline 06/2026

- Built an end-to-end DeepStream 9.0 and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction and re-streamed output.
- Converted YOLO26n to a dynamic-batch TensorRT FP16 engine on 6GB VRAM Turing hardware, validating FP16 against FP32 accuracy and profiling p99 latency to sustain 29.7 FPS per stream live.
- Implemented a **C++** and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine.
- Benchmarked IoU, NvDCF and NvSORT trackers on MOTA, IDF1, ID switches, fragmentation, FPS and VRAM to guide tracker selection.

ROS2 SLAM and Nav2 Simulation 02/2026

- Built a simulation robotics stack in ROS2 Jazzy and Gazebo Harmonic to evaluate **LiDAR**-centric SLAM, LiDAR and RGB fusion SLAM, and autonomous navigation.
- Developed ROS2 nodes, launch files and parameterised configurations in C++ and Python, using tf2 and message synchronisation to debug transform timing issues.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, plus an EKF localisation layer with robot_localization fusing wheel odometry and IMU.
- Containerised the stack with Docker and managed versioning in Git for reproducible builds.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/Modern-Edge-CV-Benchmarking

03/2026

- Benchmarked YOLOv8, YOLOv10, YOLO26, RT-DETR and EfficientViT on accuracy, latency, model size and deployment suitability for edge perception systems.
- Assessed Vision Transformer and foundation-model approaches across detection, segmentation and depth estimation.
- Exported models to ONNX and evaluated OpenVINO deployment paths against edge hardware constraints.

EDUCATION

University of Bristol — MSc Robotics, Merit

09/2023 – 09/2024 · BRISTOL, UK

Dissertation: Radiological SLAM with LiDAR Odometry

- Built a **real-time** SLAM system in **C++17** and ROS2 (roclcpp) on a Clearpath Husky, fusing Velodyne VLP-16 **LiDAR**, Intel RealSense D435i depth and radiological sensing for inspection mapping.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol.
- Configured Ceres, g2o and GTSAM as solver backends and tuned ICP parameters across environments.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering

07/2016 – 05/2020 · NAGPUR, INDIA

- Foundation in electrical and electronics engineering, underpinning later hardware and sensor-integrated systems work.

SKILLS

Languages	C++17, Python, CUDA C++, Bash, SQL
Real-time and edge	real-time C++ pipelines, multi-threaded and asynchronous processing, Intel Movidius VPU, NVIDIA Jetson AGX Xavier, Intel VTune profiling, latency optimisation
Sensors and data	multi-sensor data fusion, LiDAR (Velodyne VLP-16), depth cameras (Intel RealSense), camera and video pipelines, OpenCV, point cloud processing (PCL, Open3D)
Computer vision	NVIDIA DeepStream, GStreamer, TensorRT with custom CUDA plugins, ONNX, OpenVINO, object detection and tracking (YOLO, NvDCF)
Testing and quality	gtest, pytest, CI/CD, code review, structured logging, health checks, regression testing, Git, CMake, Linux
Robotics and SLAM	ROS2 (roclcpp, rclpy), tf2, sensor fusion, RTAB-Map, LIO-SAM, Cartographer, GTSAM, ICP and scan matching
Backend and systems	REST API design, containerised microservices, Docker, Apache Kafka, PostgreSQL, Microsoft Azure, queue-based async processing

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019)